CCCA Meta-Atlas — Cell-State Manifold Analysis

Filbin et al. 2018 · Pediatric H3-K27M mutant glioma · SmartSeq2 · 2,259 malignant cells (6 samples)

GO / NO-GO GATES

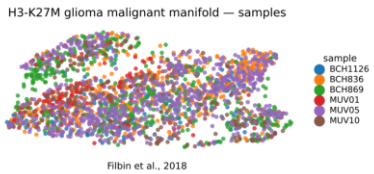
Gate 1 GO — Malignant structure is sample-mixed AND continuous. iLISI 1.00→3.81 of 6 samples after Harmony(θ=10); one connected component on the independent Harmony-PCA graph (no islands removed). Gast is not proliferation-confounded (prolif–Gast ρ=+0.09).

Gate 2 GO — Non-branching embryonic axis. Gast/pSom co-localise (joint enrichment 1.54; spatial r=+0.61) as one developmental origin; Crest is an inverse terminal gradient (pSom&Crest joint 0.22; Crest-high decile pseudotime 0.99). No Gast/pSom hub diverging into two arms — GO (non-branching).

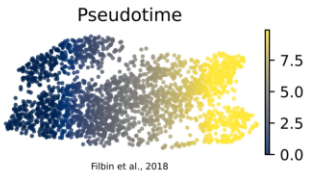
Gate 3 NA — Not applicable — no bifurcation. A single principal curve (2 tips, 0 forks) fits the manifold, so there is no branch point at which to test mutual Gast↔pSom cross-repression.

STEP FIGURES

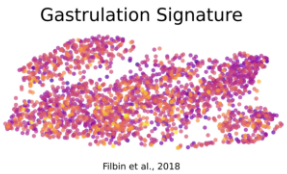
01 · Samples (integration)



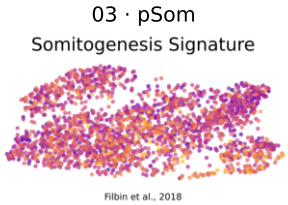
02 · Pseudotime (curve)



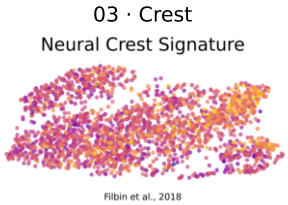
03 · Gast



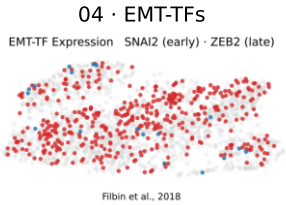
03 · pSom



03 · Crest



04 · EMT-TFs



PARAMETER CHOICES & RATIONALE

Min-cell threshold: ≥100 floor → all 6 samples kept (2,259 cells). Counts 141–667 smooth, no cliff; floor holds.

Integration: Harmony on sample, θ=10, n_pcs=20. Strong patient batch (pre-Harmony frac 0.96); plateaus by θ=5.

Geometry: scFates curve, Nodes=20 — non-branching topology (2 tips, 0 forks; self-approach 2.97 > 1).

Root node: Gast/pSom-high, Crest-low tip 2 — developmental origin (Gast/pSom early, Crest late); non-cycling.

Islands removed: 0 — single connected component (UMAP radius-graph and independent Harmony-PCA graph).

EMT-TFs shown: SNAI2 (early) · ZEB2 (late) — best-demarcating detectable pair; ZEB1/ZEB2/PRRX1 broad.

VERDICT: Pattern reproduces (non-branching). One continuous gastrulation/somitogenesis → neural-crest axis, rooted at the Gast/pSom-high origin; Crest a terminal inverse gradient. G1 GO · G2 GO · G3 NA — Linear-pSom/Crest.